

Experiment 2- TestDisk: Open-Source Data Recovery Tool

Aim :

To use TestDisk step by step to recover a missing partition and repair a corrupted one.

Installation

Linux (Debian/Ubuntu): `sudo apt-get install testdisk`

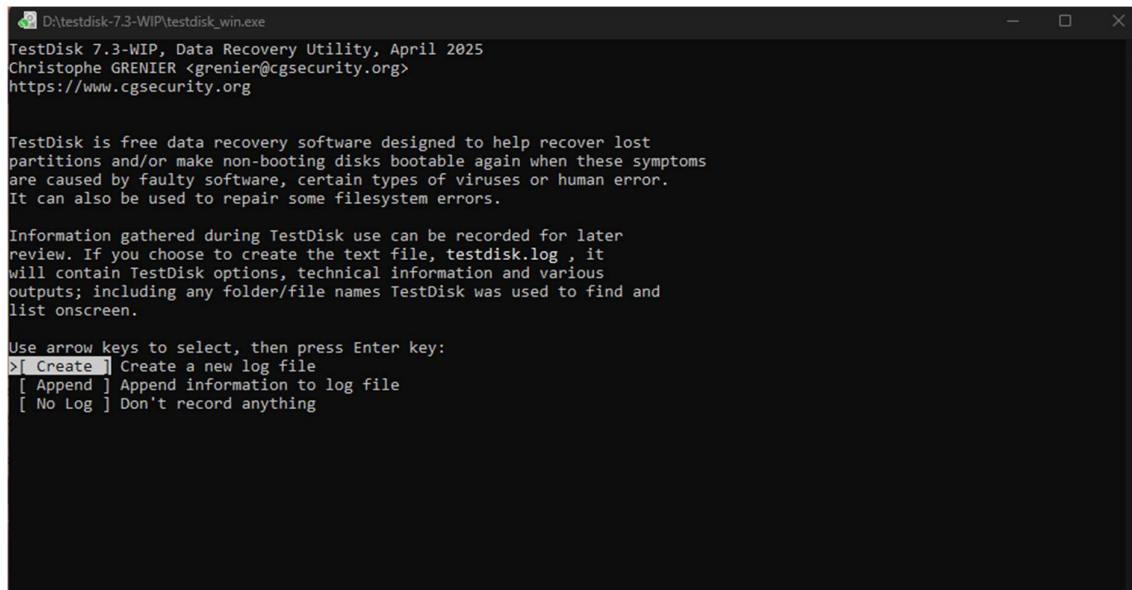
macOS (Homebrew): `brew install testdisk`

Windows: Download the executable from the official CGSecurity website.

Procedure

1. Create a Log File

- Launch TestDisk from your terminal or command prompt using `sudo testdisk` (or `testdisk_win.exe` on Windows).
- Select the [Create] option to generate a log file of the recovery session. This is helpful for future reference or troubleshooting.



```
D:\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, April 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free data recovery software designed to help recover lost
partitions and/or make non-booting disks bootable again when these symptoms
are caused by faulty software, certain types of viruses or human error.
It can also be used to repair some filesystem errors.

Information gathered during TestDisk use can be recorded for later
review. If you choose to create the text file, testdisk.log , it
will contain TestDisk options, technical information and various
outputs; including any folder/file names TestDisk was used to find and
list onscreen.

Use arrow keys to select, then press Enter key:
> [ Create ] Create a new log file
  [ Append ] Append information to log file
  [ No Log ] Don't record anything
```

2. Select the Drive to Analyze

- TestDisk will display a list of all detected storage devices.
- Use the Up/Down arrow keys to highlight the drive that contains your lost data.

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TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
>Disk \\.\PhysicalDrive0 - 1024 GB / 953 GiB - NVMe BC901 NVMe SK hynix 1024GB

>[Proceed] [Quit]

Note: Serial number 0000_0000_0000_0000_ACE4_2E00_3588_4C06.
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.
```

- Select [Proceed] to move to the next step.

3. Choose the Partition Table Type

- TestDisk will automatically suggest the most likely partition table type (e.g., Intel/PC, EFI GPT).
- The default value is usually correct. Confirm the selection by pressing Enter.

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Disk \\.\PhysicalDrive0 - 1024 GB / 953 GiB - NVMe BC901 NVMe SK hynix 1024GB

Please select the partition table type, press Enter when done.
[Intel] Intel/PC partition
>[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax] Humax partition table
[Mac] Apple partition map (legacy)
[None] Non partitioned media
[Sun] Sun Solaris partition
[XBox] Xbox partition
[Return] Return to disk selection

Hint: EFI GPT partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

4. Analyze Current Partition Structure

- From the main menu, choose [Analyse] and press Enter.

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Disk \\.\PhysicalDrive0 - 1024 GB / 953 GiB - NVMe BC901 NVMe SK hynix 1024GB
CHS 124519 255 63 - sector size=512

> [ Analyze ] Analyze current partition structure and search for lost partitions
[ Advanced ] Filesystem Utils
[ Geometry ] Change disk geometry
[ Options ] Modify options
[ Quit ] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyze'
process may give some warnings if it thinks the logical geometry is mismatched.
```

- This will display the current partition table and check for errors or missing partitions.

5. Perform a Quick Search

- After the analysis, you will be prompted to perform a [Quick Search].

```
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Disk \\.\PhysicalDrive0 - 1024 GB / 953 GiB - CHS 124519 255 63
Current partition structure:
  Partition          Start      End      Size in sectors
1 P EFI System       2048      616447  614400 [EFI system partition]
No FAT, NTFS, ext2, JFS, Reiser, cramfs or XFS marker
2 P MS Reserved      616448    878591  262144 [Microsoft reserved partition]
2 P MS Reserved      616448    878591  262144 [Microsoft reserved partition]
3 P MS Data          878592   529125375 528246784 [Basic data partition] [OS]
4 P MS Data          652005376 1302265855 650260480 [Basic data partition] [New Volume]
5 P MS Data          1302265856 1952524287 650258432 [Basic data partition] [New Volume]
6 P Windows Recovery Env 1952526336 1954553855 2027520
7 P Windows Recovery Env 1954553856 1997352959 42799104
8 P Windows Recovery Env 1997355008 2000408575 3053568
9 P EFI System       529125376 531126271 2000896
10 P Linux filesystems. data 531126272 645654527 114528256
11 P Linux Swap      645654528 652005375 6350848

P=Primary D=Deleted
> [Quick Search] [ Backup ]
Try to locate partition
```

- Select it and press Enter to scan the drive for lost partitions.
- Once a partition is found, you can press p to list its files. Deleted files and folders often appear in red.
- Press q to return to the search results.

6. Perform a Deeper Search

- If the Quick Search fails to find your lost partitions, select [Deeper Search].

```

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Disk \\.\PhysicalDrive0 - 1024 GB / 953 GiB - CHS 124519 255 63
Analyse cylinder 65862/124518: 52%

EFI System          2048      616447      614400 [EFI System Partition] [ESP]
MS Data             878592    529125368    528246777 [OS]
MS Data             878599    529125375    528246777
MS Data             529125375 1057372151    528246777

Stop

```

- This process can take a long time, as it scans the entire drive, block by block, to find remnants of partition structures.
- Again, use p to preview files and confirm if a found partition is the one you are looking for.

7. Modify Partition Status

- After finding the correct partitions, use the Left/Right arrow keys to set their status.
- Use **Left/Right arrow keys** to change status:
 - **P**: Primary
 - *****: Bootable
 - **L**: Logical
 - **D**: Deleted

```

Select D:\testdisk-7.3-WIP\testdisk_win.exe
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Disk \\.\PhysicalDrive0 - 1024 GB / 953 GiB - CHS 124519 255 63
Partition      Start      End      Size in sectors
P EFI System    2048      616447    614400 [EFI System Partition] [ESP]
D MS Data      878592    529125368    528246777 [OS]
D MS Data      878599    529125375    528246777
D MS Data      529125375 1057372151    528246777
D MS Data      652007425 1302265856    650258432
D MS Data      1302265856 1952524287    650258432 [New Volume]

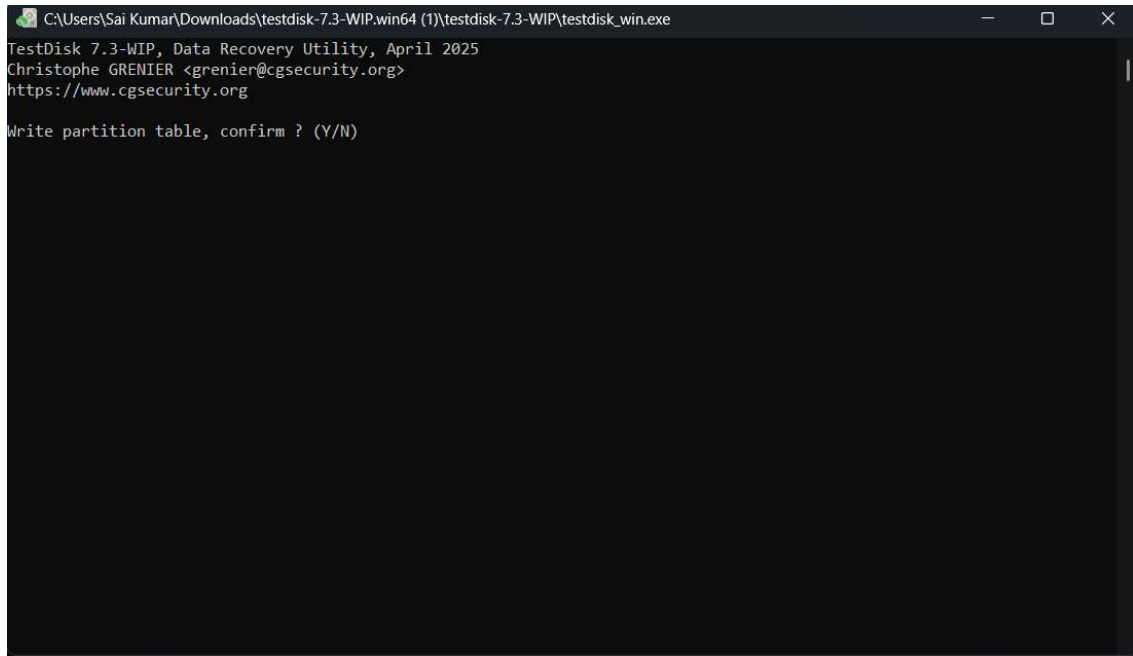
Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
P=Primary D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
FAT32, blocksize=4096, 314 MB / 300 MiB

```

- Ensure that the partitions you want to recover are marked as Primary or Logical (and not deleted).

8. Write the Partition Table

- Once you are confident the partition structure is correct, select [Write] from the menu.
- Confirm the operation by pressing y(for yes).This will write the new partition table to your disk



- WARNING: This is a permanent change. Double-check your selections before writing.

9. Recover Files

- If you just need to recover a few files without fixing the partition table, you can do so from the file list (after pressing p).
- Navigate to the folder containing your desired files.
- Use the colon : key to select the files you want to recover.
- Press the uppercase C key to copy the selected file(s).
- Navigate to a safe destination on a different storage device and press uppercase C again to paste.

10. Exit and Restart

- Once your task is complete, select [Quit] to exit the program.
- If you wrote a new partition table to the drive, it is recommended to restart your computer to allow the operating system to recognize the changes.